# Appendix B. Simple Connectivity and Jordan Curve Theorem

Equivalent forms of simple connectivity, the Jordan curve theorem, general Cauchy theorem

## Contents

**1 Equivalent formulations of simple connectivity** **2**

**2 The Jordan curve theorem** **5**

2.1 Proof of a general form of Cauchy's theorem . . . . . 9

## Overview

Simple connectivity underlies many basic results of complex analysis. Recall (Chapter 3) that a region $\Omega$ is *simply connected* if any two curves in $\Omega$ with the same end-points are homotopic; from this we deduced that

$$\int_\eta f(\zeta)\,d\zeta = 0 \tag{1}$$

for every closed curve $\eta \subset \Omega$ and every $f$ holomorphic in $\Omega$. In this appendix we prove:

(I) $\Omega$ is simply connected if and only if it is *holomorphically simply connected*, i.e. (1) holds for all closed $\eta \subset \Omega$ and all $f$ holomorphic in $\Omega$.

(II) A bounded region is simply connected if and only if its complement is connected.

(III) $\Omega$ is simply connected if and only if no curve in $\Omega$ winds around a point of $\Omega^c$.

In the second part we answer: given a simple closed curve $\gamma$, what is the region "enclosed by $\gamma$"? We prove the Jordan curve theorem for piecewise-smooth curves: $\gamma^c$ has exactly two components, the bounded one is simply connected and both have boundary $\gamma$. This yields a general form of Cauchy's theorem extending the version for toy contours of Chapter 2. As in Chapter 1, "curve" means "piecewise-smooth curve" unless stated otherwise.

## 1 Equivalent formulations of simple connectivity

**Theorem 1.1**

A region $\Omega$ is holomorphically simply connected if and only if $\Omega$ is simply connected.

**Proof.** If $\Omega$ is simply connected, it is holomorphically simply connected by Corollary 5.3. Conversely, let $\Omega$ be holomorphically simply connected. If $\Omega = \mathbb{C}$ it is simply connected. Otherwise the proof of the Riemann mapping theorem (Theorem 3.1) applies verbatim: the only use of simple connectivity there is to construct a holomorphic logarithm (hence square root) of a non-vanishing holomorphic function $g$, namely $\log g(z) = c + \int_{z_0}^z g'/g$, which is well defined precisely because (1) holds. Hence $\Omega$ is conformally equivalent to $\mathbb{D}$. Since $\mathbb{D}$ is simply connected and a conformal map carries homotopies to homotopies, $\Omega$ is simply connected. ∎

**Theorem 1.2**

If $\Omega$ is a bounded region in $\mathbb{C}$, then $\Omega$ is simply connected if and only if $\Omega^c$ is connected.

Boundedness is needed: an infinite strip is simply connected but its complement has two components. If complements are taken in the Riemann sphere, the conclusion holds for all regions.

**Proof that $\Omega^c$ connected $\Rightarrow$ $\Omega$ simply connected.** By Theorem 1.1 it suffices to show $\int_\eta f = 0$ for $\eta \subset \Omega$ closed and $f$ holomorphic in $\Omega$. With $d(z, \Omega^c) = \inf\{|z - w| : w \in \Omega^c\}$, let

$$K_\delta = \{z \in \overline{\Omega} : d(z, \Omega^c) \geq \delta\}.$$

$K_\delta$ is compact ($\Omega$ is bounded), $K_\delta \subset \Omega$, and $\eta \subset K_\delta$ for small $\delta$ since $\eta$ is compact. We show that $K_\delta^c$ is connected.

Suppose $K_\delta^c = O_1 \cup O_2$ with $O_1, O_2$ disjoint, open, non-empty, and set $F_i = O_i \cap \Omega^c$. Since $\Omega^c \subset K_\delta^c$, $\Omega^c = F_1 \cup F_2$ with $F_1 \cap F_2 = \emptyset$.

*Step $F_1$ is closed.* Let $z_n \in F_1$, $z_n \to z$. Then $z \in \Omega^c$ ($\Omega^c$ is closed), so $z \in O_1 \cup O_2$. If $z \in O_2$, then $z_n \in O_2$ for large $n$ as $O_2$ is open, contradicting $z_n \in O_1$. So $z \in O_1$, i.e. $z \in F_1$. Likewise $F_2$ is closed.

*Step* $F_1 \neq \emptyset$. Otherwise $\Omega^c \subset O_2$. Pick $w \in O_1$. Since $w \notin K_\delta$, there is $z \in \Omega^c$ with $|w - z| < \delta$. Every point of the segment $[w, z]$ is within distance $< \delta$ of $z \in \Omega^c$, hence lies in $K_\delta^c$. So the connected segment lies in $O_1 \cup O_2$ and meets both ($w \in O_1$, $z \in O_2$), a contradiction. Likewise $F_2 \neq \emptyset$.

Thus $\Omega^c$ is disconnected, contrary to hypothesis; so $K_\delta^c$ is connected. By Runge's theorem (Theorem 5.7) there are polynomials $P_n \to f$ uniformly on $K_\delta$, hence on $\eta$. Since $\int_\eta P_n = 0$ (polynomials have primitives), $\int_\eta f = \lim \int_\eta P_n = 0$. ∎

The converse is proved after Theorem 1.4, using winding numbers.

## Winding numbers

If $\gamma$ is closed and $z \notin \gamma$, each loop of $\gamma$ around $z$ changes $\frac{1}{2\pi}\arg(\zeta - z)$ by $\pm 1$. Since $\log w = \log|w| + i \arg w$, the quantity $\frac{1}{2\pi i}\int_\gamma \frac{d\zeta}{\zeta - z}$ should count these loops.

**Definition (Winding number)**

The **winding number** (or **index**) of a closed curve $\gamma$ around $z \notin \gamma$ is

$$W_\gamma(z) = \frac{1}{2\pi i}\int_\gamma \frac{d\zeta}{\zeta - z}.$$

For $\gamma(t) = e^{ikt}$, $0 \le t \le 2\pi$ ($k \in \mathbb{N}$), $W_\gamma(z) = k$ for $|z| < 1$ and 0 for $|z| > 1$; for $e^{-ikt}$ the values are $-k$ and 0. For a positively oriented toy contour, $W_\gamma = 1$ in its interior and 0 in its exterior.

**Lemma 1.3**

Let $\gamma$ be a closed curve in $\mathbb{C}$.

(i) If $z \notin \gamma$, then $W_\gamma(z) \in \mathbb{Z}$.

(ii) If $z, w$ lie in the same connected component of $\gamma^c$, then $W_\gamma(z) = W_\gamma(w)$.

(iii) If $z$ lies in the unbounded component of $\gamma^c$, then $W_\gamma(z) = 0$.

**Proof.** (i) Let $\gamma : [0,1] \to \mathbb{C}$ parametrize the curve and $G(t) = \int_0^t \frac{\gamma'(s)}{\gamma(s) - z}\, ds$. $G$ is continuous and, except at finitely many points, differentiable with $G' = \gamma'/(\gamma - z)$. Then $H(t) = (\gamma(t) - z)e^{-G(t)}$ is continuous with $H' = \gamma' e^{-G} - (\gamma - z)G'e^{-G} = 0$ except at finitely many points, so $H$ is constant: $H(1) = H(0) = \gamma(0) - z$. As $\gamma(1) = \gamma(0) \neq z$, $e^{-G(1)} = 1$, so $G(1) = 2\pi i\, W_\gamma(z) \in 2\pi i\mathbb{Z}$.

(ii) $W_\gamma$ is a continuous function of $z \in \gamma^c$ (the integrand depends continuously on $z$, uniformly on $\gamma$), integer-valued by (i), hence constant on each connected component.

(iii) $|W_\gamma(z)| \le \frac{\ell(\gamma)}{2\pi\, d(z,\gamma)} \to 0$ as $|z| \to \infty$; since $W_\gamma$ is constant on the unbounded component by (ii), it is 0 there. ∎

**Theorem 1.4**

A bounded region $\Omega$ is simply connected if and only if $W_\gamma(z) = 0$ for every closed curve $\gamma \subset \Omega$ and every $z \notin \Omega$.

**Proof.** If $\Omega$ is simply connected and $z \notin \Omega$, then $f(\zeta) = 1/(\zeta - z)$ is holomorphic in $\Omega$, and Cauchy's theorem gives $W_\gamma(z) = 0$.

Conversely, by the part of Theorem 1.2 already proved, it suffices to show $\Omega^c$ is connected. Suppose not: $\Omega^c = F_1 \cup F_2$ with $F_1, F_2$ disjoint, closed, non-empty. Only one of them can be unbounded ($\Omega$ is bounded, so $\Omega^c$ contains the connected set $\{|z| > R\}$), so we may assume $F_1$ is bounded, hence compact. We construct a closed curve $\gamma \subset \Omega$ and $w \in F_1$ with $W_\gamma(w) \neq 0$, using the following

lemma. ■

**Lemma 1.5**

Under the above assumptions, let $w \in F_1$. There is a finite collection $\mathcal{Q} = \{Q_1, \dots, Q_n\}$ of closed squares of a uniform grid $\mathcal{G}$ such that:

(i) $w$ is in the interior of $Q_1$;

(ii) the interiors of $Q_j$ and $Q_k$ are disjoint for $j \neq k$;

(iii) $F_1$ is contained in the interior of $\bigcup_j Q_j$;

(iv) $\bigcup_j Q_j$ is disjoint from $F_2$;

(v) the boundary of $\bigcup_j Q_j$ lies in $\Omega$ and is a finite union of disjoint simple closed polygonal curves.

**Proof of Theorem 1.4, assuming Lemma 1.5.** Orient each $\partial Q_j$ positively. Since $w$ is interior to $Q_1$ and $w \notin Q_j$ for $j > 1$,

$$\sum_{j=1}^{n} \frac{1}{2\pi i} \int_{\partial Q_j} \frac{d\zeta}{\zeta - w} = 1. \tag{2}$$

A side shared by two squares of $\mathcal{Q}$ is traversed twice in opposite directions and cancels. The remaining sides, oriented with $\bigcup Q_j$ on the left, form the polygons $\gamma_1, \dots, \gamma_M$ of (v), so $\sum_{j=1}^{M} W_{\gamma_j}(w) = 1$. Hence $W_{\gamma_{j_0}}(w) \neq 0$ for some $j_0$, while $\gamma_{j_0} \subset \Omega$ and $w \notin \Omega$: a contradiction. ■

**Proof of Lemma 1.5.** $F_1$ is compact and $F_2$ closed, so $d = d(F_1, F_2) > 0$. Take a grid $\mathcal{G}_0$ of closed squares of side $\delta < d/100$ such that $w$ is the center of a square $R_1$, and let $\mathcal{R} = \{R_1, \dots, R_m\}$ be the squares of $\mathcal{G}_0$ meeting $F_1$.

*Step (i)–(iv) and $\partial \bigcup R_j \subset \Omega$.* (i), (ii) are clear. Every point of $\bigcup R_j$ is within $\sqrt{2}\delta < d$ of $F_1$, giving (iv). If $z \in F_1$, every square of $\mathcal{G}_0$ containing $z$ meets $F_1$ and so belongs to $\mathcal{R}$; these squares cover a neighborhood of $z$, giving (iii). A *boundary side* (a side of exactly one square of $\mathcal{R}$) is a side of a square not in $\mathcal{R}$, so it misses $F_1$; it also misses $F_2$ by (iv). Hence $\partial \bigcup R_j \subset \Omega$.

*Step Bad vertices.* The boundary of $\bigcup R_j$ is the union of the boundary sides; their end-points are boundary vertices. At a grid vertex $P$ the four surrounding squares give 0, 2 or 4 boundary sides at $P$, and 4 occurs exactly when two diagonally opposite squares are in $\mathcal{R}$ and the other two are not. Call such a $P$ *bad*.

*Step Refinement.* Let $\mathcal{G}$ be the grid obtained by cutting each square of $\mathcal{G}_0$ into 9 equal subsquares, and take all $9m$ subsquares of squares of $\mathcal{R}$, with $Q_1$ the middle subsquare of $R_1$ (so $w$ is its center). At each bad vertex $P$ add the two subsquares of $\mathcal{G}$ having $P$ as a corner and lying in the two missing squares of $\mathcal{G}_0$. Call the result $\mathcal{Q} = \{Q_1, \dots, Q_n\}$. Now all four small squares at $P$ are in $\mathcal{Q}$, so $P$ is interior. An added subsquare $s$ at $P$ has three other corners. At the two lying on sides of $\mathcal{G}_0$, three of the four surrounding small squares are in $\mathcal{Q}$. At the one inside a big square, only $s$ is. So none of them is bad. Added squares at different bad vertices sit at different corners of a big square, so they do not touch. Hence $\mathcal{Q}$ has no bad vertices. (i)–(iv) persist: the added squares lie in squares of $\mathcal{G}_0$ adjacent to $\mathcal{R}$, within $3\delta < d$ of $F_1$. The new boundary sides lie in squares of $\mathcal{G}_0$ not in $\mathcal{R}$, so they miss $F_1$, and also $F_2$. So $\partial \bigcup Q_j \subset \Omega$.

*Step (v).* Every boundary vertex of $\bigcup Q_j$ now meets exactly two boundary sides. With the orientation leaving $\bigcup Q_j$ on the left, one of them enters the vertex and the other leaves it (check the cases of one, two adjacent, or three surrounding squares). Starting from a boundary side $[a_1, a_2]$ we get a unique continuation $[a_2, a_3], [a_3, a_4], \dots$. There are finitely many sides, so some vertex repeats; let $a_m$ be the first repetition, $a_m = a_k$ with $k < m$. If $k > 1$, then $a_k$ meets the three distinct sides $[a_{k-1}, a_k]$, $[a_k, a_{k+1}]$, $[a_{m-1}, a_m]$, which is impossible. So $k = 1$ and $a_1 \cdots a_m$ is a simple closed polygon. Removing its sides and repeating exhausts the boundary sides, giving finitely many disjoint simple closed polygons (disjoint because each vertex lies on only one of them). ■

**Proof of Theorem 1.2, converse.** Let $\Omega$ be bounded and simply connected. If $\Omega^c$ were disconnected, the proof of Theorem 1.4 produces a closed curve $\gamma \subset \Omega$ and $w \notin \Omega$ with $W_\gamma(w) \neq 0$. This contradicts the first part of Theorem 1.4, since $\Omega$ is simply connected. ■

## 2 The Jordan curve theorem

The curves in the statements are piecewise-smooth, but the proofs also use curves that are merely continuous (the curves $\gamma_\eta$ below).

**Theorem 2.1 (Jordan arc theorem)**

Let $\gamma$ be a simple (non-closed) piecewise-smooth curve in the plane. Then $\gamma^c$ is an open connected set whose boundary is precisely $\gamma$.

**Theorem 2.2 (Jordan curve theorem)**

Let $\gamma$ be a simple closed piecewise-smooth curve in the plane. Then $\gamma^c$ consists of two disjoint connected open sets. Exactly one of them is bounded and simply connected; it is the **interior** $\Omega_\gamma$ of $\gamma$. The other is unbounded; it is the **exterior** $\mathcal{U}_\gamma$. With the appropriate orientation of $\gamma$,

$$W_\gamma(z) = \begin{cases} 1 & z \in \Omega_\gamma, \\ 0 & z \in \mathcal{U}_\gamma. \end{cases}$$

**Remark**

Both theorems hold for merely continuous curves, but the proofs are harder; the piecewise-smooth case suffices for many applications.

**Theorem 2.3**

If $f$ is holomorphic in the interior $\Omega_\gamma$ of a simple closed curve $\gamma$, then $\int_\eta f(\zeta)\,d\zeta = 0$ for every closed curve $\eta \subset \Omega_\gamma$.

**Proof (supplied).** $\Omega_\gamma$ is a simply connected region by Theorem 2.2, so this is Cauchy's theorem, Corollary 5.3. ■

**Idea of the proof of Theorem 2.1.** $\gamma^c$ is open, so it suffices to show it is pathwise connected. Join $z, w \notin \gamma$ by a curve. If it meets $\gamma$, first go from $z$ to a point $z'$ and from $w$ to a point $w'$ near $\gamma$ without meeting $\gamma$. Then join $z'$ to $w'$ by travelling "parallel" to $\gamma$, going around an end-point if necessary. If $\gamma$ parametrizes a smooth piece, $\gamma'(t) \neq 0$ is tangent to it, $i\gamma'(t)$ is normal, and $\gamma(t) + i\eta\gamma'(t)$ is a "parallel" curve.

In the next three lemmas and two propositions, $\gamma_0$ is a *simple smooth* curve with arc-length parametrization $\gamma : [0, L] \to \mathbb{C}$, $|\gamma'(t)| = 1$ (every smooth curve has one). An *interior point* of $\gamma_0$ is a point $\gamma(t)$ with $0 < t < L$ (not to be confused with the interior of Theorem 2.2).

**Lemma 2.4**

For $\eta \in \mathbb{R}$ let $\gamma_\eta$ be the continuous curve $\gamma_\eta(t) = \gamma(t) + i\eta\gamma'(t)$, $0 \le t \le L$. There is $\eta_1 > 0$ such that $\gamma_0 \cap \gamma_\eta = \emptyset$ whenever $0 < |\eta| < \eta_1$.

**Proof.**

*Step Local version.* For $s, t \in [0, L]$,

$$\gamma_\eta(t) - \gamma(s) = \int_s^t \gamma'(u)\,du + i\eta\gamma'(t) = \int_s^t [\gamma'(u) - \gamma'(t)]\,du + (t - s + i\eta)\gamma'(t).$$

By uniform continuity of $\gamma'$ there is $\delta > 0$ with $|\gamma'(x) - \gamma'(y)| < 1/2$ for $|x - y| < \delta$. If $|s - t| < \delta$, then using $|\gamma'(t)| = 1$ and $2|a + ib| \geq |a| + |b|$,

$$|\gamma_\eta(t) - \gamma(s)| \geq |t - s + i\eta| - \frac{|t - s|}{2} \geq \frac{|\eta|}{2}.$$

So $\gamma_\eta(t) \neq \gamma(s)$ when $|t - s| < \delta$ and $\eta \neq 0$.

*Step Global version.* Choose $0 = t_0 < \cdots < t_n = L$ with $t_{k+1} - t_k < \delta/2$, and set $I_k = \{t : |t - t_k| \leq \delta/4\}$, $J_k = \{t : |t - t_k| < \delta/2\}$, $J_k' = \{t \in [0, L] : |t - t_k| \geq \delta/2\}$ (intersected with $[0, L]$). The $I_k$ cover $[0, L]$. If $t \in I_k$ and $s \in J_k$ then $|t - s| < \delta$, so

$$\gamma_\eta(I_k) \cap \gamma(J_k) = \emptyset \qquad (\eta \neq 0). \tag{3}$$

Since $\gamma_0$ is simple, the compact sets $\gamma(I_k)$ and $\gamma(J_k')$ are disjoint, at distance $d_k > 0$. For $t \in I_k$, $s \in J_k'$,

$$|\gamma_\eta(t) - \gamma(s)| \geq |\gamma(t) - \gamma(s)| - |\eta| \geq d_k - |\eta| \geq d_k/2 \quad \text{if } |\eta| \leq d_k/2,$$

so

$$\gamma_\eta(I_k) \cap \gamma(J_k') = \emptyset \qquad (|\eta| < d_k/2). \tag{4}$$

With $\eta_1 = \min_k d_k/2$, (3) and (4) give $\gamma_0 \cap \gamma_\eta = \emptyset$ for $0 < |\eta| < \eta_1$. ■

**Lemma 2.5**

Suppose $z \notin \gamma_0$ and $z_0 \in \gamma_0$ is a point closest to $z$ with $z_0 = \gamma(t_0)$, $0 < t_0 < L$ (this holds if $z$ is closer to an interior point of $\gamma_0$ than to either end-point). Then $z = \gamma(t_0) + i\eta\gamma'(t_0)$ for some real $\eta \neq 0$, i.e. $z \in \gamma_\eta$.

**Proof.** Near $t_0$, $z - \gamma(t) = z - \gamma(t_0) - \gamma'(t_0)(t - t_0) + o(|t - t_0|)$, so

$$|z - z_0|^2 \leq |z - \gamma(t)|^2 = |z - z_0|^2 - 2(t - t_0)\,\mathrm{Re}\,[(z - \gamma(t_0))\overline{\gamma'(t_0)}] + o(|t - t_0|).$$

Since $t - t_0$ takes both signs, $\mathrm{Re}[(z - \gamma(t_0))\overline{\gamma'(t_0)}] = 0$, i.e. $(z - \gamma(t_0))\overline{\gamma'(t_0)} = i\eta$ with $\eta \in \mathbb{R}$. As $|\gamma'(t_0)| = 1$, $\overline{\gamma'(t_0)} = 1/\gamma'(t_0)$, so $z - \gamma(t_0) = i\eta\gamma'(t_0)$; and $\eta \neq 0$ because $z \notin \gamma_0$. ■

If $z \in \gamma_\eta$ and $w \in \gamma_\epsilon$ ($\eta, \epsilon \neq 0$) are obtained as in Lemma 2.5, we say $z, w$ are on the **same side** of $\gamma_0$ if $\eta, \epsilon$ have the same sign, and on **opposite sides** otherwise. We do not define "the two sides of $\gamma_0$"; the notion refers to the chosen representations, and we do not claim the two cases are exclusive. Points on the same side can be joined almost directly by a curve parallel to $\gamma_0$. Points on opposite sides require going around an end-point.

**Proposition 2.6**

Let $A, B$ be the end-points of $\gamma_0$ and $K$ a compact set with $\gamma_0 \cap K = \emptyset$ or $\gamma_0 \cap K = \{A, B\}$ (or a subset thereof). If $z, w \notin \gamma_0$ lie on the same side of $\gamma_0$, and each is closer to an interior point of $\gamma_0$ than to $K$ or to the end-points of $\gamma_0$, then $z$ and $w$ can be joined by a continuous curve in the complement of $K \cup \gamma_0$.

**Proof.** By Lemma 2.5, with $z_0 = \gamma(t_0)$, $w_0 = \gamma(s_0)$ interior points closest to $z, w$,

$$z = \gamma(t_0) + i\eta_0\gamma'(t_0), \qquad w = \gamma(s_0) + i\epsilon_0\gamma'(s_0),$$

with $\eta_0, \epsilon_0$ of the same sign, say positive, and say $t_0 \leq s_0$.

*Step Normal segments.* The half-open segment from $z$ to $z_0$ lies in $(K \cup \gamma_0)^c$. If $p$ is on it with $p \neq z_0$ and $p \in K \cup \gamma_0$, then $|z-p| < |z-z_0| = d(z, K \cup \gamma_0)$, which is absurd. So for all small $\eta > 0$, $z$ can be joined to $z_\eta = \gamma(t_0) + i\eta\gamma'(t_0)$, and likewise $w$ to $w_\eta = \gamma(s_0) + i\eta\gamma'(s_0)$.

*Step Parallel curve.* Choose $\eta < \eta_1$ (Lemma 2.4) and $\eta$ smaller than the distance from $K$ to $\{\gamma(t) : t_0 \le t \le s_0\}$. This distance is positive since this arc avoids $A, B$. Then $\{\gamma_\eta(t) : t_0 \le t \le s_0\}$ joins $z_\eta$ to $w_\eta$. It misses $\gamma_0$ by Lemma 2.4, and it misses $K$ since $|\gamma_\eta(t) - \gamma(t)| = \eta$. ■

**Lemma 2.7**

There is $\eta_2 > 0$ such that the half-neighborhood

$$N = \{\gamma(L) + \eta e^{i\theta}\gamma'(L) : -\pi/2 \le \theta \le \pi/2,\ 0 < \eta < \eta_2\}$$

is disjoint from $\gamma_0$.

**Proof.** As in Lemma 2.4,

$$\gamma(L) + \eta e^{i\theta}\gamma'(L) - \gamma(t) = \int_t^L [\gamma'(u) - \gamma'(L)]\,du + (L - t + \eta e^{i\theta})\gamma'(L).$$

Choose $\delta$ with $|\gamma'(u) - \gamma'(L)| < 1/2$ for $|u - L| < \delta$. Since $\cos\theta \ge 0$, $|L - t + \eta e^{i\theta}|^2 \ge (L-t)^2 + \eta^2 \ge \frac{1}{2}(L - t + \eta)^2$. So for $L - \delta < t \le L$,

$$|\gamma(L) + \eta e^{i\theta}\gamma'(L) - \gamma(t)| \ge \frac{L - t + \eta}{\sqrt{2}} - \frac{L-t}{2} \ge \frac{\eta}{\sqrt{2}} > 0.$$

Thus $\gamma(t) \notin N$ for $L - \delta < t \le L$. Finally, take $\eta_2$ less than the distance from $\gamma(L)$ to $\{\gamma(t) : 0 \le t \le L - \delta\}$, which is positive since $\gamma_0$ is simple. ■

**Proposition 2.8**

Let $A$ be an end-point of $\gamma_0$ and $K$ a compact set with $\gamma_0 \cap K = \emptyset$ or $\gamma_0 \cap K = \{A\}$. If $z, w \notin \gamma_0$ are closer to interior points of $\gamma_0$ than to $K$ or to the end-points of $\gamma_0$, then $z$ and $w$ can be joined by a continuous curve in the complement of $\gamma_0 \cup K$.

**Proof.** If $z, w$ are on the same side, apply Proposition 2.6. So let them be on opposite sides, and (reversing the parametrization if needed) $A = \gamma(0)$, so the other end-point $B = \gamma(L) \notin K$. Write

$$z = \gamma(t_0) + i\eta_0\gamma'(t_0), \qquad w = \gamma(s_0) - i\epsilon_0\gamma'(s_0), \qquad \eta_0, \epsilon_0 > 0,$$

and let $\tau = \min(t_0, s_0) > 0$. Choose $\eta > 0$ smaller than $\eta_1$, $\eta_2$ (Lemmas 2.4, 2.7), and the distance from $K$ to $\{\gamma(t) : \tau \le t \le L\}$ (positive since this arc avoids $A$). As in Proposition 2.6, join $z$ to $\gamma_\eta(t_0)$ by a normal segment. Then follow $\gamma_\eta$ from $t_0$ to $L$ to reach $\gamma(L) + i\eta\gamma'(L)$. Then follow the half-circle $\gamma(L) + \eta e^{i\theta}\gamma'(L)$, $\theta$ from $\pi/2$ down to $-\pi/2$, inside $N$, to reach $\gamma(L) - i\eta\gamma'(L)$. Then follow $\gamma_{-\eta}$ from $L$ back to $s_0$, and the normal segment to $w$. The parallel pieces miss $\gamma_0$ by Lemma 2.4; the half-circle misses $\gamma_0$ by Lemma 2.7. All pieces lie within $\eta$ of $\{\gamma(t) : \tau \le t \le L\}$, so they miss $K$. ■

**Proof of Theorem 2.1.**

*Step Boundary.* $O = \gamma^c$ is open and $\partial O \subset \gamma$. Every point of a smooth piece of $\gamma$ is a limit of points of $O$: by Lemma 2.4 (applied to a small sub-arc, and noting the rest of $\gamma$ is at positive distance), $\gamma(t) + i\eta\gamma'(t) \notin \gamma$ for small $\eta \neq 0$. Since $\partial O$ is closed and the smooth points are dense in $\gamma$, $\partial O = \gamma$.

*Step Connectedness: induction on the number of smooth pieces.* First let $\gamma$ be smooth, with end-points $A, B$, and let $Z, W \notin \gamma$. Join $Z$ to $W$ by a polygonal path $\Lambda$ avoiding $A$ and $B$. If $\Lambda$ misses $\gamma$ we are done. Otherwise let $p$ and $q$ be the first and last points of $\Lambda$ on $\gamma$; they are interior points of $\gamma$. Points $z$ of $\Lambda$ just before $p$ satisfy $d(z, \gamma) \le |z - p| \to 0$, while $|z - A|, |z - B|$ stay bounded

below. So we may take such a $z \notin \gamma$ closer to an interior point of $\gamma$ than to $A, B$, joined to $Z$ along $\Lambda$ without meeting $\gamma$. Similarly take $w$ just after $q$. Proposition 2.8 with $K = \emptyset$ joins $z$ to $w$ in $\gamma^c$.

Now suppose the result holds for curves of $n-1$ smooth pieces, and let $\gamma = K \cup \gamma_0$ with $K$ the union of $n-1$ consecutive pieces and $\gamma_0$ smooth. Then $K$ is compact and $K \cap \gamma_0$ is a single end-point $A$ of $\gamma_0$. Given $Z, W \notin \gamma$, the induction hypothesis gives a curve from $Z$ to $W$ missing $K$. Since $K^c$ is open and connected, we may take it polygonal and avoiding the other end-point of $\gamma_0$. If it meets $\gamma_0$, it does so at interior points. As above, take $z$ (resp. $w$) just before the first (resp. after the last) meeting point; it is closer to an interior point of $\gamma_0$ than to $K$ or to the end-points, since its distance to $K$ is bounded below. Proposition 2.8 joins $z$ to $w$ in $(\gamma_0 \cup K)^c = \gamma^c$. ∎

**Proof of Theorem 2.2.**

*Step At most two components.* Fix $W$ outside a large disc containing $\gamma$, and let $\mathcal{U}$ be the set of points joined to $W$ by a continuous curve in $\gamma^c$. $\mathcal{U}$ is open and connected, and contains the exterior of the disc. Let $\Omega = \gamma^c - \mathcal{U}$; we show $\Omega$ is connected. Remove a smooth piece $\gamma_0$ (with end-points $A, B$) from $\gamma$, and let $K$ be the remaining arc, so $K \cap \gamma_0 = \{A, B\}$. For $Z \in \Omega$, Theorem 2.1 applied to $K$ gives a curve $\Lambda_Z$ from $Z$ to $W$ missing $K$. Since $Z \notin \mathcal{U}$, $\Lambda_Z$ meets $\gamma_0$, necessarily at interior points. As in the proof of Theorem 2.1, choose $z, w \in \Lambda_Z$ closer to interior points of $\gamma_0$ than to $K$ or to $A, B$, such that the pieces of $\Lambda_Z$ from $Z$ to $z$ and from $w$ to $W$ lie in $\gamma^c$. Then $z, w$ are on opposite sides of $\gamma_0$. Otherwise Proposition 2.6 would join $z$ to $w$ in $\gamma^c$, hence $Z$ to $W$, contradicting $Z \notin \mathcal{U}$. If $Z_1 \in \Omega$ is another point, with corresponding $z_1, w_1$ on opposite sides, then $z$ and $z_1$ are on the same side. Otherwise $z_1$ is on the same side as $w$, and Proposition 2.6 joins $Z_1$ to $W$ in $\gamma^c$, a contradiction. So Proposition 2.6 joins $z$ to $z_1$ in $\gamma^c$, and $Z$, $Z_1$ lie in the same component. Thus $\gamma^c$ has at most two components, $\mathcal{U}$ and (if non-empty) $\Omega$.

*Step Exactly two components.* By Lemma 1.3 it suffices to find points with different winding numbers. Parametrize $\gamma$ and fix $z_0 = \gamma(t_0)$ interior to a smooth piece, $v = \gamma'(t_0) \neq 0$. For $\epsilon > 0$ let

$$z_\epsilon = \gamma(t_0) + i\epsilon v, \qquad w_\epsilon = \gamma(t_0) - i\epsilon v.$$

For small $\epsilon$ these avoid $\gamma$, and the normal segments $\{\gamma(t_0) \pm isv : 0 < s \leq \epsilon\}$ lie in $\gamma^c$ (Lemma 2.4 for the piece, plus the positive distance from $\gamma(t_0)$ to the rest of $\gamma$). By Lemma 1.3(ii),

$$\Delta = |W_\gamma(z_\epsilon) - W_\gamma(w_\epsilon)|$$

is a constant integer for all small $\epsilon > 0$. With $a(t) = \gamma(t) - \gamma(t_0)$,

$$\frac{\gamma'(t)}{\gamma(t) - z_\epsilon} - \frac{\gamma'(t)}{\gamma(t) - w_\epsilon} = \frac{2i\epsilon v\, \gamma'(t)}{a(t)^2 + \epsilon^2 v^2}.$$

Since $a(t) = v(t - t_0) + o(|t - t_0|)$ and $\gamma'(t) = v + o(1)$,

$$\frac{2i\epsilon v \gamma'(t)}{a(t)^2 + \epsilon^2 v^2} = \frac{2i\epsilon}{(t - t_0)^2 + \epsilon^2}\,(1 + \rho(t, \epsilon)),$$

where, given $\sigma > 0$, there is $\delta > 0$ with $|\rho(t, \epsilon)| < \sigma$ for $|t - t_0| \leq \delta$, uniformly in $\epsilon$. (Indeed $a^2 + \epsilon^2 v^2 = v^2[(t - t_0)^2 + \epsilon^2] + o((t - t_0)^2)$ and $o((t - t_0)^2)/[(t - t_0)^2 + \epsilon^2] \to 0$.) Split the integral defining $W_\gamma(z_\epsilon) - W_\gamma(w_\epsilon)$ into $|t - t_0| \geq \delta$ and $|t - t_0| < \delta$. On the first part $\gamma(t)$ stays at positive distance from $\gamma(t_0)$, while $z_\epsilon, w_\epsilon \to \gamma(t_0)$, so this part tends to 0 as $\epsilon \to 0$. On the second,

$$\frac{1}{2\pi i} \int_{|t - t_0| < \delta} \frac{2i\epsilon\, dt}{(t - t_0)^2 + \epsilon^2} = \frac{1}{\pi} \int_{-\delta/\epsilon}^{\delta/\epsilon} \frac{ds}{s^2 + 1} = \frac{2}{\pi} \arctan(\delta/\epsilon) \to 1,$$

and the $\rho$-term is at most $\frac{\sigma}{\pi} \int_{\mathbb{R}} \frac{ds}{s^2+1} = \sigma$ in absolute value. Letting $\epsilon \to 0$ gives $|\Delta - 1| \leq \sigma$ for every $\sigma > 0$, so $\Delta = 1$.

Hence $z_\epsilon$ and $w_\epsilon$ lie in different components, so $\gamma^c$ has exactly the two components $\mathcal{U}$ and $\Omega$. $\mathcal{U}$ is unbounded with $W_\gamma = 0$ there (Lemma 1.3(iii)), and $\Omega$ is bounded. Since $\Delta = 1$, $W_\gamma = \pm 1$ on $\Omega$,

and after possibly reversing orientation $W_\gamma = 1$ there. Every smooth point of $\gamma$ is a limit of the $z_\epsilon$ and of the $w_\epsilon$, hence of points of both components. Smooth points are dense in $\gamma$, and the boundary of each component lies in $\gamma$. So $\gamma = \partial\Omega = \partial\mathcal{U}$. Write $\Omega_\gamma = \Omega$ and $\mathcal{U}_\gamma = \mathcal{U}$.

*Step $\Omega_\gamma$ is simply connected.* By Theorem 1.2 it suffices that $\Omega_\gamma^c = \gamma \cup \mathcal{U}_\gamma$ is connected. Suppose $\Omega_\gamma^c = F_1 \cup F_2$ with $F_1, F_2$ closed, disjoint, non-empty, and let $O_i = \mathcal{U}_\gamma \cap F_i$. The sets $O_1, O_2$ are disjoint and $\mathcal{U}_\gamma = O_1 \cup O_2$.

$O_1$ is open. Let $z \in O_1$; small balls $B$ at $z$ lie in $\mathcal{U}_\gamma \subset F_1 \cup F_2$. If every such ball met $F_2$, then $z \in F_2$ ($F_2$ closed), contradicting $F_1 \cap F_2 = \emptyset$. So some ball lies in $F_1$, hence in $O_1$. Likewise $O_2$ is open.

$O_1 \neq \emptyset$. Otherwise $F_1 \subset \gamma$ and $\mathcal{U}_\gamma \subset F_2$. Take $z \in F_1 \subset \gamma$. Since $\gamma = \partial\mathcal{U}_\gamma$, every ball at $z$ meets $\mathcal{U}_\gamma \subset F_2$, so $z \in F_2$, a contradiction. Likewise $O_2 \neq \emptyset$.

This disconnects $\mathcal{U}_\gamma$, a contradiction. This completes the proof of the Jordan curve theorem for piecewise-smooth curves. ■

## 2.1 Proof of a general form of Cauchy's theorem

**Theorem 2.9**

If $f$ is holomorphic in an open set containing a simple closed piecewise-smooth curve $\gamma$ and its interior, then $\int_\gamma f = 0$.

Let $\mathcal{O} \supset \gamma \cup \Omega_\gamma$ be open with $f$ holomorphic in $\mathcal{O}$. The idea is to build a closed curve $\Gamma \subset \Omega_\gamma$ so close to $\gamma$ that $\int_\gamma f = \int_\Gamma f$. The right side vanishes by Theorem 2.3, since $f$ is holomorphic in the simply connected set $\Omega_\gamma$. Near smooth parts of $\gamma$, $\Gamma$ is essentially a parallel curve $\gamma_\eta$ (Lemma 2.4); near the junctions of smooth parts, $\Gamma$ follows arcs of circles.

**Lemma 2.10**

Let $\gamma : [0,1] \to \mathbb{C}$ be a simple smooth curve. For all sufficiently small $\epsilon > 0$ the circle $C_\epsilon$ of center $\gamma(0)$ and radius $\epsilon$ meets $\gamma$ in exactly one point.

**Proof.** Assume $\gamma(0) = 0$. Since $\gamma(1) \neq 0$, the intermediate value theorem applied to $|\gamma(t)|$ gives at least one intersection point for $\epsilon < |\gamma(1)|$. Suppose the claim fails: there are $\epsilon_j \to 0$ such that $|\gamma(t)| = \epsilon_j$ has two distinct solutions. Since $|\gamma(t)| \geq c(\tau) > 0$ for $t \geq \tau$ (simplicity), both solutions tend to 0. By the mean value theorem applied to $h(t) = |\gamma(t)|^2$, there are $t_j \to 0$, $t_j > 0$, with $h'(t_j) = 2\,\mathrm{Re}[\gamma'(t_j)\overline{\gamma(t_j)}] = 0$. But $\gamma(t) = \gamma'(0)t + t\varphi(t)$ and $\gamma'(t) = \gamma'(0) + \psi(t)$ with $\varphi, \psi \to 0$ as $t \to 0$, so

$$\mathrm{Re}[\gamma'(t)\overline{\gamma(t)}] = |\gamma'(0)|^2 t + o(t),$$

which is $\neq 0$ for small $t > 0$ since $\gamma'(0) \neq 0$. This is a contradiction. ■

**Proof of Theorem 2.9.** Choose $\epsilon$ so small that the open set $U$ of points at distance $< \epsilon$ from $\gamma$ is contained in $\mathcal{O}$. Let $P_1, \ldots, P_n$ be the consecutive junction points of the smooth parts of $\gamma$. Choose $\delta < \epsilon/10$ so small that each circle $C_j$ centered at $P_j$ of radius $\delta$ meets $\gamma$ in exactly two points. This is possible by Lemma 2.10, applied to the two smooth parts meeting at $P_j$; the rest of $\gamma$ is at positive distance from $P_j$. The two points determine two arcs of $C_j$, exactly one of which, $\mathcal{C}_j$, has its interior in $\Omega_\gamma$. Indeed, for a smooth part with end-point $P_j$ and small $\eta$, the curves $\gamma_\eta$ and $\gamma_{-\eta}$ lie on opposite sides of $\gamma$, hence in different components of $\gamma^c$ (proof of Theorem 2.2), and both meet $C_j$. The disc $D_j$ of center $P_j$ and radius $2\delta$ lies in $U$, hence in $\mathcal{O}$.

Let $B_j$ be the end-point of $\mathcal{C}_j$ on the smooth part $\gamma_j$ from $P_j$ to $P_{j+1}$, and $A_{j+1}$ the end-point of $\mathcal{C}_{j+1}$ on $\gamma_j$. Take a chain of discs $\mathcal{D} = \{D_0, \ldots, D_K\}$ in $U$ covering $\gamma$, with $D_k \cap D_{k+1} \neq \emptyset$, $D_0 = D_K$, and with the discs $D_j$ among them. By Lemma 2.4 there is a continuous curve $\Gamma_j$ in $\Omega_\gamma$ and in the

union of the discs, running parallel to $\gamma_j$, from a point near $B_j$ on $\mathcal{C}_j$ to a point near $A_{j+1}$ on $\mathcal{C}_{j+1}$. Since $\gamma$ is only $C^1$, $\Gamma_j$ need not be smooth; replacing it by a nearby polygonal line we may assume it is. Joining $\Gamma_j$ to $\Gamma_{j+1}$ along a piece of $\mathcal{C}_{j+1}$, and so on, gives a closed piecewise-smooth curve $\Gamma \subset \Omega_\gamma$.

$f$ has a primitive on each disc of $\mathcal{D}$ (Theorem 2.1). Arguing as in the proof of Theorem 5.1, and comparing $\gamma$ and $\Gamma$ disc by disc, gives $\int_\gamma f = \int_\Gamma f$: on consecutive discs the primitives differ by constants, and the endpoint contributions telescope. Since $\Gamma \subset \Omega_\gamma$ and $\Omega_\gamma$ is simply connected, $\int_\Gamma f = 0$ by Theorem 2.3. Hence $\int_\gamma f = 0$. ∎